UNIT QUESTIONS

SKILLS CRITICAL THINKING 4

1

a Which of the following statements is incorrect?

A Gravitational forces keep our Moon in orbit around the Earth.

B Comets and moons orbit planets.

C Planets orbit the Sun.

D Artificial satellites orbit a planet. **(1)**

SKILLS REASONING 3

b Which of the following sequences might be followed by a star that has a mass much larger than our Sun?

A main sequence, stellar nebula, red giant, white dwarf

B stellar nebula, main sequence, red supergiant, black hole

C stellar nebula, main sequence, white dwarf, black hole

D main sequence, red giant, neutron star, white dwarf **(1)**

SKILLS CRITICAL THINKING 5

c Which are the most common reactions that take place in the core of a star that is in the main sequence?

A nuclear fission reactions between helium nuclei

B nuclear fission reactions between hydrogen nuclei

C nuclear fusion reactions between helium nuclei

D nuclear fusion reactions between hydrogen nuclei **(1)**

d Which of the following observations does not support the Big Bang theory?

A cosmic microwave background radiation

B blue-shift in the spectra of galaxies

C red-shift in the spectra of galaxies

D accelerating galaxies **(1)**

(Total for Question 1 = 4 marks)

6

2

Copy and complete the following passage about gravitational forces, filling in the spaces.

Gravitational forces cause planets and _____ to _____ the Sun. The planets _____ to the Sun, for example, _____ and _____, experience the _____ forces and so have the most _____ orbits. The planets _____ from the Sun, for example, _____ and _____, experience the _____ forces and so have the least _____ orbits. The shape of a planet's orbit is not quite _____. It is _____.

All objects that orbit a planet are called _____ but natural ones are called_____.

(Total for Question 2 = 16 marks)

SKILLS PROBLEM SOLVING, ANALYSIS 5

3

A man has a mass of 80kg. On the Earth he weighs 800N. Using the table on page 263 calculate the weight of this man on the surface of:

a Venus **(3)**

b Mars **(3)**

c Neptune. **(3)**

(Total for Question 3 = 9 marks)

SKILLS PROBLEM SOLVING 9

4 Calculate the speed of a satellite that is orbiting 250 km above the Earth's surface and completes one orbit in 2 hr.

(Total for Question 4 = 5 marks)

SKILLS ANALYSIS

5 The table below contains information about some of the planets in our Solar System.

Planet	Surface gravity compared with the Earth	Distance from Sun compared with the Earth	Period/Earth years
Mercury	0.4	0.4	0.2
Venus	0.9	0.7	0.6
Mars	0.4	1.5	1.9
Jupiter	2.6	5.0	12
Saturn	1.1	9.5	30

4

a Name three planets that have a weaker gravitational pull on their surface than there is on Earth. **(1)**

HINT

circumference of a circle = $2\pi r$

b How long is a year on Saturn? **(1)**

SKILLS PROBLEM SOLVING 5

c If the distance from the Earth to the Sun is 150 million kilometers, calculate the distance of Saturn from the Sun. **(2)**

9

d Assuming that the orbital path for Saturn is circular, calculate its orbital speed in km/s. **(3)**

(Total for Question 5 = 7 marks)

SKILLS INTERPRETATION 6

6 **a** Draw a diagram to show the shape of the orbit of a comet around the Sun. **(2)**

SKILLS REASONING 7

b Describe and explain how the speed of a comet changes as it travels around its orbit. **(4)**

(Total for Question 6 = 6 marks)

SKILLS INTERPRETATION 6

7 Draw the Hertzsprung–Russell diagram. Label the main features and explain why the diagram is useful for astronomers.

((Total for Question 7 = 12 marks)

5

8 What is cosmic microwave background radiation?

(Total for Question 8 = 3 marks)

SKILLS PROBLEM SOLVING 8

9 The spectra of two galaxies, A and B, each show a change in the position of absorption lines of 200 nm and 450 nm compared with the Sun's spectrum. If these lines correspond to light with a wavelength of 500 nm in the Sun's spectrum calculate the speed of these galaxies relative to the speed of the Earth. (speed of light = 3×10^8 m/s)

(Total for Question 9 = 10 marks)